LLM

Hello learners, have you heard about the term Large Language Model or LLM? Well, it's quite a buzzword in AI world. In this video, let's understand some basic fundamentals about LLMs.

- **-Define Large Language Model, LLM.**
- **-Differentiate between general purpose, specialized and open source LLMs and their business applications.**
- **-Identify the pros and cons of LLM.**
- **-List the applications of LLMs in business.**

Let's start simple.

Imagine you are typing a WhatsApp message and your phone suggests the next word "meeting" today or tomorrow. That's a small-scale language model at work. Now, scale that up a million times with far more data, smarter predictions, and creative abilities. That's what we call a Large Language Model or LLM.

So, what exactly is an LLM?

An LLM is a type of artificial intelligence trained on massive amounts of text from books, articles, websites, and even open source code. It learns patterns in language and uses them to predict and generate words, sentences, and even entire documents. Think of it as a super-intelligent writing partner who has read almost everything on the internet.

LLMs come in many flavors.

- **General-purpose models like OpenAI's GPT-4 or Google's Jmini which can chat, answer questions, or write essays.**
- **Specialized models like GitHub's Copilot, built to assist coders, or healthcare-focused models trained on medical data.**
- **Open-source models like Hugging Faces Bloom or MetaSlama, where communities contribute to development and training.**

Each type serves different business needs, from drafting emails to analyzing contracts or even predicting customer behavior.

Why are LLMs significant?

For managers, the significance lies in impact.

LLMs can automate repetitive tasks, augment human decision-making, and spark creativity at scale.

Consider Morgan Stanley, which uses GPT-powered tools to help financial advisors quickly pull insights from decades of research reports. Or KPMG, which uses LLMs for tax and compliance automation.

These aren't futuristic dreams, they're happening now.

Let us now discuss the pros and cons of LLMs.

<u>pros of LLMs</u>

- **Productivity boost : Drafting reports, summarizing meetings, or generating presentations can be cut down from hours to minutes.**
- **Scalability : They handle millions of queries simultaneously like customer chatbots in global businesses.**
- **Multilingual capability: Because they're trained on global text, they can work across languages critical for multinational firms.**

<u>cons of LLMs.</u>

- **Hallucination:**

  Sometimes, LLMs make up facts. For example, ChatGPT might confidently cite a study that doesn't exist.

- **Bias:**

  If the training data carries stereotypes, the model can reproduce them.

- **Resource heavy:**

  Training and running these models requires enormous computing power, making them costly and energy intensive. That's why companies like PwC are investing in governance frameworks, deciding when and how LLMs should be trusted.

Applications of LLM in business.

So, where do managers see the value? A few big areas stand out.

- **Content creation: Marketing teams use Jasper or Writesonic, both powered by LLMs, to generate ad, copy, blogs, or product descriptions at scale.**
- **Customer support: Banks and e-commerce firms are deploying LLM-driven chatbots to resolve queries in real time, freeing up human agents for complex issues.**
- **Knowledge Management: Retrieval Augmented Generation lets companies build private, chat GPTs on their internal data. Imagine asking your firm's AI what were last year's top three customer complaints and getting an instant answer.**
- **Code Assistance: Developers use GitHub Copilot powered by LLMs to autocomplete code and reduce errors.**
- **Decision Support: From risk assessment in insurance to deal structuring in M&A, LLMs are helping leaders explore scenarios faster than ever.**

To summarize, for managers, LLMs are not just about technology, they're about transformation. They change how teams work, how decisions are made, and how quickly organizations can adapt. Like any powerful tool, they need thoughtful use, balancing efficiency with accuracy, and innovation with responsibility. So next time you hear the term LLM, don't think of it as just another tech buzzword. Think of it as a strategic lever, one that can help you lead smarter, faster, and with more creativity in today's competitive world.

Understanding LLM hallucinations.

- **-Describe why LLMs hallucinate.**
- **-Evaluate the risks hallucinations pose to trust, compliance, and reputation in different industries.**
- **-Determine the practical ways to manage hallucinations. Let's start with a simple question.**

Have you ever spoken with an AI tool like ChatGPT and thought, wow, that sounds right, only to realize later it was completely wrong? That's what we call a hallucination in large language models, or LLMs.

<u>What is LLM Hallucination?</u>

Hallucination means the AI gives an answer that looks convincing, sometimes even detailed, but is factually incorrect or irrelevant.

Imagine asking an AI for a book recommendation, and it confidently invents a title with an ISBN number that doesn't exist. To you, it feels real. To the AI, it's just predicting words that fit together.

<u>types of hallucinations.</u>

There are a few common ways hallucinations show up.

- **<u>Sentence contradictions:</u> The AI says, "The sky is blue today," and then, in the same response, "The sky is gray with heavy clouds." Contradictory answers confuse rather than clarify.**
- **<u>Logical fallacies:</u> Like describing a man holding a pitcher in one hand and a basket in the other, while also clapping, forgetting that humans only have two hands.**
- **<u>Prompt contradictions:</u> You ask for a positive restaurant review, and it says, the food was terrible. The response ignores your instructions.**
- **<u>Factual errors:</u> Saying, Barack Obama was the first US president. Sounds confident, but it's plainly wrong. Irrelevant generations. You ask, what country's capital is Paris? It answers correctly with France, but then adds a**

random fact about a singer named Paris, which has nothing to do with the question.

Why large language models hallucinate?

Unlike calculators that always give one correct answer, LLMs are trained to predict the most likely next word based on patterns in massive amounts of text. Sometimes there isn't just one right answer.

For example, translating a sentence from English to Bengali, there may be several valid translations. Writing a letter from a daughter to her late father, there's no single correct way to do it. This flexibility makes LLMs powerful for creativity, but it also opens the door to hallucinations.

On top of that, issues like data quality, bias, insufficient training examples, or even the way the AI handles long texts add to the problem.

managerial implications of LLM hallucinations:

For business leaders, hallucinations are not just a technical quirk, they're a strategic concern.

If your customer-facing chatbot invents refund policies or your internal AI assistant cites fake legal precedents, the risks are real. Trust, compliance, and reputation are at stake.

But it's not all negative. Knowing hallucinations exist helps managers set the right expectations.

LLMs are storytellers, not knowledge bases. They're great at drafting, summarizing, and ideating, but they need human oversight for accuracy.

examples of hallucinations.

- **Amazon and product listings: AI tools have been caught inventing fake product reviews or ISBN numbers raising trust issues.**
- **Healthcare: An AI tool once generated treatment guidelines that looked credible but were factually dangerous, highlighting why human experts must validate outputs.**
- **A US law. Firm faced penalties after lawyers submitted AI generated citations that turned out to be fake. A costly reminder of why hallucinations matter at the boardroom level.**

different ways to manage LLM hallucinations.

While we can't eliminate hallucinations entirely, managers can mitigate risks by:

- **Using AI with guardrails: Many companies now build private with retrieval augmented generation , so responses come from verified company data rather than the open internet.**
- **Cross-checking AI output: Encourage teams to validate important results just like: Fact-checking a consultant's report**
- **Training teams: Make sure staff understand AI's strengths and weaknesses. Treat it as a smart assistant, not an all-knowing oracle.**

To summarize: So, if there's one thing to remember, LLMs hallucinate because they're designed to generate plausible language, not guaranteed facts. For managers, this means using them wisely, leveraging their creativity and speed, but always adding a layer of human judgment.

In short, hallucinations don't make LLMs useless. They make them tools that need thoughtful handling. As leaders, understanding this difference will help you harness Gen AI for innovation without being blindsided by its limitations.

how to measure the extent of hallucinations and how to minimize those.

Large language models, LLMs, can occasionally generate false information, known as hallucinations. The question is, how do we measure them, and more importantly, how do we reduce them, so businesses can trust AI in real-world use?

measuring AI hallucinations.

Think of measuring hallucinations like grading a student's answer. You can check for word overlap, meaning, or relevance.

lexical checks: tools like Blue or Rouge compare AI's answers, word for word, against a correct answer. This is like marking spelling and sentence match. They work well for translations or summaries.

Example, in machine translation, Rouge can tell whether the sky is clear.. is close enough to the sky is blue.

Semantic similarity: Here, we focus on meaning, not just words. Metrics like BERT score or BLERT check whether the AI's output makes logical sense compared to references.

Example, if the AI cites a law that doesn't exist, semantic checks flag it as suspicious.

Self-verification: This is cutting edge. The AI double-checks itself using methods like self-check GPT or RAGAS. These methods work even without a fixed, ground truth, making them practical for business contexts.

Example, an AI customer service bot answering queries about refund policy could verify its answer against your company's database.

So, measuring hallucinations is not perfect. Different industries use different yardsticks, but it's becoming an essential layer of AI quality control.

So, **how do we minimize hallucinations,** those AI-generated inaccuracies that can disrupt trust and decision-making?

1. **adjusting AI settings**. Tweaking parameters like the temperature controls how creative or conservative the AI is.

For example, a banking chatbot explaining loan terms should be cautious, sticking closely to facts instead of generating imaginative answers. This reduces the chance of misleading customers.

2. retrieval augmented generation or RAG:

RAG grounds AI's responses in verified, company-specific data rather than generic internet content.

Take Morgan Stanley, they developed an AI tool for financial advisors that pulls answers solely from their trusted research reports. This way, the information is accurate and tailored to their business needs.

3. step-by-step fact-checking, sometimes called chain of verification:

Here the AI first drafts a response, then systematically checks each part for correctness before delivering the final answer. Imagine asking the AI which politicians were born in New York. Chain of verification ensures only verified names appear, preventing fabricated details.

4. Guardrails and monitoring:

Platforms like NVIDIA's Nemo guardrails allow businesses to set clear boundaries for AI systems defining what's acceptable for the AI to say, what's off limits, and when human review is necessary. This real-time oversight can stop hallucinations before they reach customers.

5. having the AI show where its answers come from.
Think of it like adding footnotes. For example, if a pharma company uses an AI assistant, it could point directly to a section of a research paper or a regulatory guideline. That way, the user isn't just taking the AI's word for it, they can check the source and feel confident the information is accurate.

Why should managers care deeply about hallucinations?

Because they're not just technical glitches, they represent serious business risks.
Picture

- a law, firm AI citing cases that don't exist,
- a hospital AI recommending unsafe treatments,
- or a telecom chatbot providing incorrect billing information.

Each error erodes trust, damages reputation, and risks compliance violations. Hallucinations are not going away completely, but they can be measured, managed, and reduced.

The right approach depends on context. A creative AI blog writer can afford some imagination, but a financial AI advisor cannot.
So as managers, your role is not to become data scientists, but to ask the right questions.
-How is hallucination measured in our use case?
-What guardrails are in place?
-And when do humans step in?